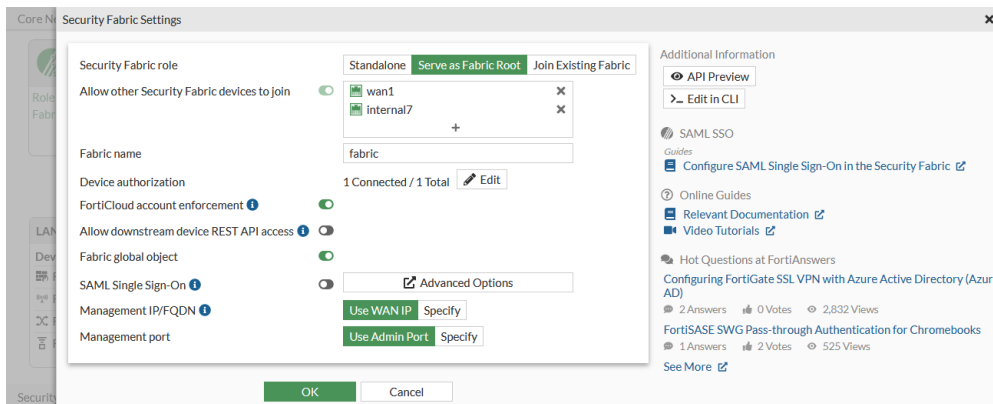
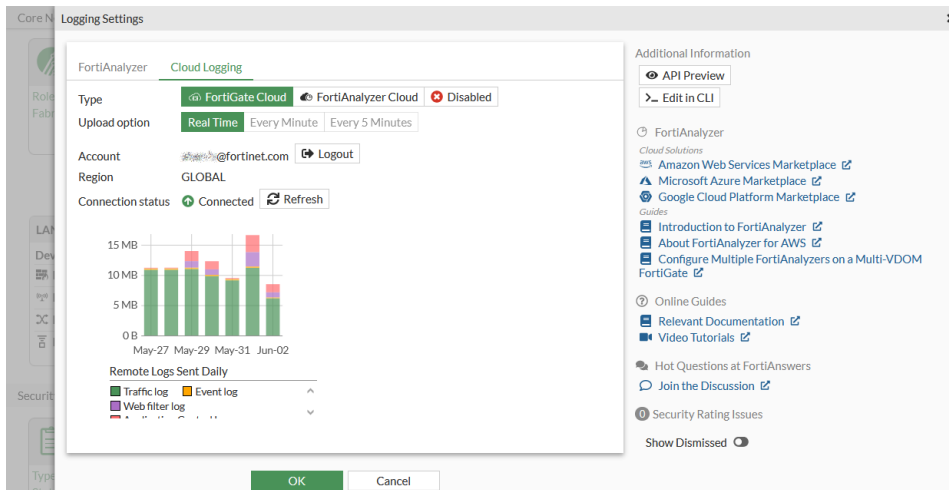




2. Configure FortiCloud logging on the root FortiGate:

- a. Go to *Security Fabric > Fabric Connectors* and double-click the *Logging & Analytics* card.
- b. On the *Cloud Logging* tab, set *Type* to *FortiGate Cloud*.



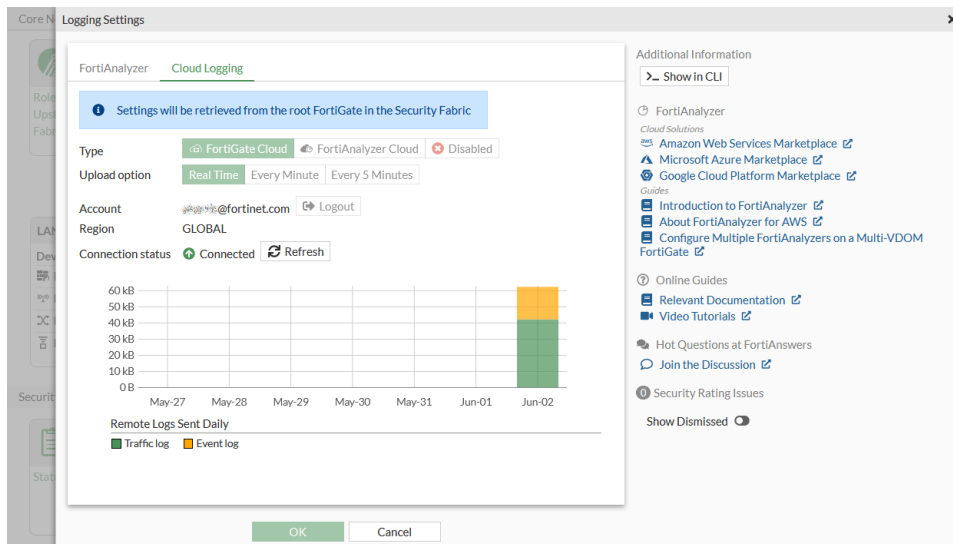
- c. Click **OK**.

3. Configure the FGT-F-VM to join the Security Fabric:

- a. Go to *Security Fabric > Fabric Connectors* and double-click the *Security Fabric Setup* card.
- b. Set *Security Fabric role* to *Join Existing Fabric*.
- c. Click **OK**. The FortiGate is authorized and successfully joins the Security Fabric.

4. Check the FortiCloud logging settings:

- a. Go to *Security Fabric > Fabric Connectors* and double-click the *Logging & Analytics* card.
- b. Go to the *Cloud Logging* tab. The settings are automatically retrieved from the root FortiGate and the *Account* is the same.



To configure a Security Fabric with FortiCloud logging in the CLI:

```
config log fortiguard setting
    set status enable
    set upload-option realtime
end
```

The FortiCloud account enforcement setting is enabled by default in the Security Fabric settings:

```
show system csf
config system csf
    set status enable
    set group-name "CSF_101"
    set forticloud-account-enforcement enable
end
```

Cloud sandboxing

FortiGate Cloud can be used for automated sample tracking, or sandboxing, for files from a FortiGate. This allows suspicious files to be sent to be inspected without risking network security. If the file exhibits risky behavior, or is found to contain a virus, a new virus signature is created and added to the FortiGuard antivirus signature database.

See [Configuring sandboxing on page 4574](#) for instructions to configure FortiGate Cloud Sandbox. Sandboxing results are shown on the *Sandbox* tab in the FortiGate Cloud portal.

Configuring FortiAnalyzer Cloud

FortiAnalyzer Cloud differs from FortiAnalyzer in the following ways:

- You cannot enable FortiAnalyzer Cloud in vdom override-setting when global FortiAnalyzer Cloud is disabled.
- You must use the CLI to retrieve and display logs sent to FortiAnalyzer Cloud. The FortiOS GUI is not supported.

- You cannot enable FortiAnalyzer Cloud and FortiGate Cloud at the same time.

For more information, see [Licensing](#) in the FortiAnalyzer Cloud Deployment Guide.

In the *Security Fabric > Fabric Connectors > Logging & Analytics* card settings, *FortiAnalyzer Cloud* is grayed out when you do not have a FortiAnalyzer Cloud entitlement. When you have a FortiAnalyzer Cloud entitlement, *FortiAnalyzer Cloud* is available and you can authenticate by the certificate.

In FortiAnalyzer Cloud, you can view logs from FortiOS in the *Event > All Types* page.

#	Date/Time	Level	Device ID	Action	Message	User	User Interface
1	10:52:45	alert	FGSH1E5800000000		Configuration is changed in the admin session	admin	ssh(10.6.30.254)
2	05-01-18:07	alert	FGSH1E5800000000		Configuration is changed in the admin session	admin	ssh(10.6.30.254)
3	05-01-18:00	alert	FGSH1E5800000000		Configuration is changed in the admin session	admin	ssh(10.6.30.254)
4	05-01-17:57	alert	FGSH1E5800000000	login	Administrator ddd login failed from https(10.6.30.254)	ddd	https(10.6.30.254)
5	05-01-17:57	information	FGSH1E5800000000	Edit	Edit log.fortianalyzer-cloud.filter	admin	ssh(10.6.30.254)
6	05-01-17:56	information	FGSH1E5800000000	Edit	Edit log.setting	admin	ssh(10.6.30.254)
7	05-01-17:56	notice	FGSH1E5800000000	connect	Connected to FortiAnalyzer.fortianalyzer.for...		
8	05-01-17:55	alert	FGSH1E5800000000	login	Administrator ccc login failed from https(10.6.30.254)	ccc	https(10.6.30.254)
9	05-01-17:55	alert	FGSH1E5800000000	login	Administrator bbb login failed from https(10.6.30.254)	bbb	https(10.6.30.254)
10	05-01-17:53	alert	FGSH1E5800000000	login	Administrator aaa login failed from https(10.6.30.254)	aaa	https(10.6.30.254)
11	05-01-17:53	information	FGSH1E5800000000	Edit	Edit log.fortianalyzer-cloud.override-filter	admin	ssh(10.6.30.254)
12	05-01-17:53	information	FGSH1E5800000000	logout	Administrator admin timed out on https(10.6.30.254)		https(10.6.30.254)
13	05-01-17:53	notice	FGSH1E5800000000	perf-stats	Performance statistics: average CPU: 0, mem...		
14	05-01-17:51	notice	FGSH1E5800000000	Delete	Delete 1 old report files		
15	05-01-17:51	notice	FGSH1E5800000000	connect	Connected to FortiAnalyzer.fortianalyzer.for...		
16	05-01-17:48	notice	FGSH1E5800000000	perf-stats	Performance statistics: average CPU: 0, mem...		
17	05-01-17:48	information	FGSH1E5800000000	Delete	Delete 1 old report files		
18	05-01-17:48	information	FGSH1E5800000000	Delete	Delete 2 old report files		
19	05-01-17:45	information	FGSH1E5800000000	login	Administrator admin logged in successfully fr...	admin	https(10.6.30.254)
20	05-01-17:45	notice	FGSH1E5800000000	connect	Connected to FortiAnalyzer.fortianalyzer.for...		
21	05-01-17:33	information	FGSH1E5800000000	Delete	Delete 1 old report files		
22	05-01-17:21	information	FGSH1E5800000000	Edit	Edit log.setting	admin	ssh(10.6.30.254)
23	05-01-17:20	information	FGSH1E5800000000	login	Administrator admin logged in successfully fr...	admin	https(10.6.30.254)
24	05-01-17:20	information	FGSH1E5800000000	login	Administrator admin logged in successfully fr...	admin	ssh(10.6.30.254)
25	05-01-17:20	information	FGSH1E5800000000	connect	Connected to FortiAnalyzer.fortianalyzer.for...		
26	05-01-17:20	notice	FGSH1E5800000000	connect	Connected to FortiAnalyzer.fortianalyzer.for...		
27	05-01-17:18	information	FGSH1E5800000000	Edit	Edit system.admin.admin	admin	GUI(10.6.30.254)
28	05-01-17:18	information	FGSH1E5800000000	Edit	Edit log.fortianalyzer-cloud.setting	admin	GUI(10.6.30.254)
29	05-01-17:18	notice	FGSH1E5800000000	connect	Connected to FortiAnalyzer.fortianalyzer.for...		
30	05-01-17:16	information	FGSH1E5800000000	login	Administrator admin logged in successfully fr...	admin	https(10.6.30.254)
31	05-01-17:14	notice	FGSH1E5800000000	login	The http daemon adjusted time from Web Ma...	FortiKli:FS2Z4D3Z14000736	

To configure FortiAnalyzer Cloud logging in the GUI:

- Go to *Security Fabric > Fabric Connectors* and double-click the *Logging & Analytics* card.
- Select the *Cloud Logging* tab and set the *Type* to *FortiAnalyzer Cloud*.
- Optionally, configure the remaining log settings:

Upload option

Select the frequency of log uploads to the remote device:

- Real Time:** logs are sent to the remote device in real time.
- Every Minute:** logs are sent to the remote device once every minute. This option is unavailable if the Security Fabric connection is configured.
- Every 5 Minutes:** logs are sent to the remote device once every five minutes. This is the default option. This option is unavailable if the Security Fabric connection is configured.

Allow access to FortiGate REST API

Define access to FortiGate REST API:

- Enable:** the REST API accesses the FortiGate topology and shares data and results.
- Disable:** the REST API does not share data and results.

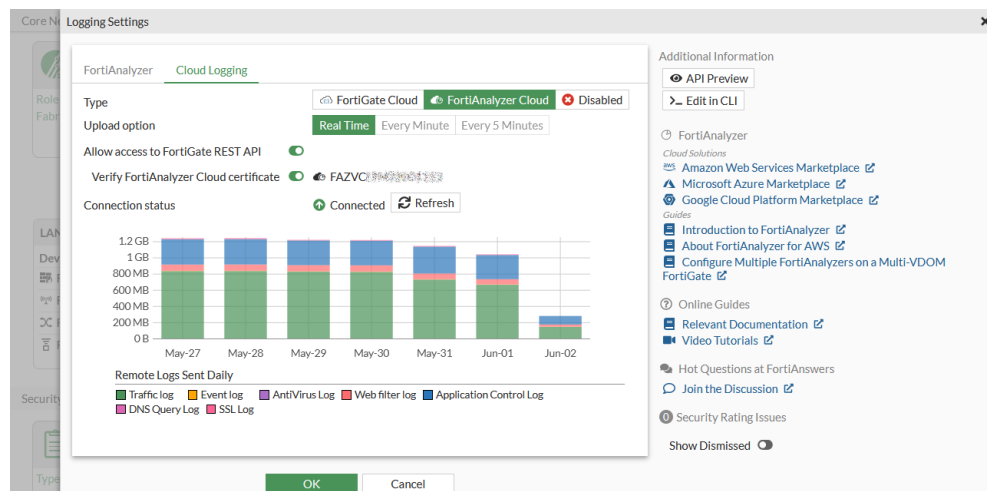
Verify FortiAnalyzer Cloud certificate

Define the FortiAnalyzer Cloud certificate verification process:

- Enable:** the FortiGate will verify the FortiAnalyzer Cloud serial number against the FortiAnalyzer certificate. When verified, the serial number is stored in the FortiGate configuration.
- Disable:** the FortiGate will not verify the FortiAnalyzer Cloud certificate against the serial number.

- Click **OK**. A prompt appears to verify the FortiAnalyzer Cloud serial number.

5. Click *Accept*.
6. The verified FortiAnalyzer Cloud certificate appears in the settings.



To enable FortiAnalyzer Cloud logging in the CLI:

1. Configure the FortiAnalyzer Cloud settings:

```
config log fortianalyzer-cloud setting
    set status enable
    set ips-archive disable
    set certificate-verification enable
    set serial "FAZVCLTM19000000"
    set access-config enable
    set enc-algorithm high
    set ssl-min-protocol-version default
    set conn-timeout 10
    set monitor-keepalive-period 5
    set monitor-failure-retry-period 5
    set upload-option realtime
end
```

2. Configure the FortiAnalyzer Cloud filters:

```
config log fortianalyzer-cloud filter
    set severity information
    set forward-traffic disable
    set local-traffic disable
    set multicast-traffic disable
    set sniffer-traffic disable
    set anomaly disable
    set voip disable
    set dlp-archive disable
end
```

To disable FortiAnalyzer Cloud logging for a specific VDOM in the CLI:

1. Enable override FortiAnalyzer in the general log settings:

```
config log setting
    set faz-override enable
end
```

2. Disable the override FortiAnalyzer Cloud setting:

```
config log fortianalyzer-cloud override-setting
    set status disable
end
```

To set FortiAnalyzer Cloud logging to filter for a specific VDOM in the CLI:

1. Enable override FortiAnalyzer in the general log settings:

```
config log setting
    set faz-override enable
end
```

2. Enable the override FortiAnalyzer Cloud setting:

```
config log fortianalyzer-cloud override-setting
    set status enable
end
```

3. Configure the override filters for FortiAnalyzer Cloud:

```
config log fortianalyzer-cloud override-filter
    set severity information
    set forward-traffic disable
    set local-traffic disable
    set multicast-traffic disable
    set sniffer-traffic disable
    set anomaly disable
    set voip disable
    set dlp-archive disable
end
```

To display FortiAnalyzer Cloud logs in the CLI:

```
# execute log filter device fortianalyzer-cloud
# execute log filter category event
# execute log display
```

Sample log

```
date=2019-05-01 time=17:57:45 idseq=60796052214644736 bid=100926 dvid=1027 itime="2019-05-01
17:57:48" euid=3 epid=3 dsteuid=0 dstepid=3 logver=602000890 logid=0100032002 type="event"
```

```
subtype="system" level="alert" srcip=10.6.30.254 dstip=10.6.30.9 action="login" msg="Administrator
ddd login failed from https(10.6.30.254) because of invalid user name" logdesc="Admin login
failed" sn="0" user="ddd" ui="https(10.6.30.254)" status="failed" reason="name_invalid"
method="https" eventtime=1556758666274548325 devid="FG5H1E5818900000" vd="root" dtime="2019-05-01
17:57:45" itime_t=1556758668 devname="FortiGate-501E"
```

```
date=2019-05-01 time=17:57:21 idseq=60796052214644736 bid=100926 dvid=1027 itime="2019-05-01
17:57:23" euid=3 epid=3 dsteuid=0 dstepid=3 logver=602000890 logid=0100044546 type="event"
subtype="system" level="information" action="Edit" msg="Edit log.fortianalyzer-cloud.filter "
logdesc="Attribute configured" user="admin" ui="ssh(10.6.30.254)" cfgtid=164757536
cfgpath="log.fortianalyzer-cloud.filter" cfgattr="severity[information->critical]"
eventtime=1556758642413367644 devid="FG5H1E5818900000" vd="root" dtime="2019-05-01 17:57:21"
itime_t=1556758643 devname="FortiGate-501E"
```

Configuring FortiClient EMS

The FortiGate Security Fabric root device can link to FortiClient Endpoint Management System (EMS) and FortiClient EMS Cloud (a cloud-based EMS solution) for endpoint connectors and automation. Multiple EMS servers can be added to the Security Fabric, including FortiClient EMS Cloud server. EMS settings are synchronized between all Fabric members.

FortiGates with FortiCare Premium or FortiCare Elite have 3 free seats to FortiClient EMS Cloud. See [FortiCloud Subscriptions](#) for details.

To enable cloud-based EMS services, the FortiGate must be registered to FortiCloud with an appropriate user account. The following examples presume that the EMS certificate has already been configured.

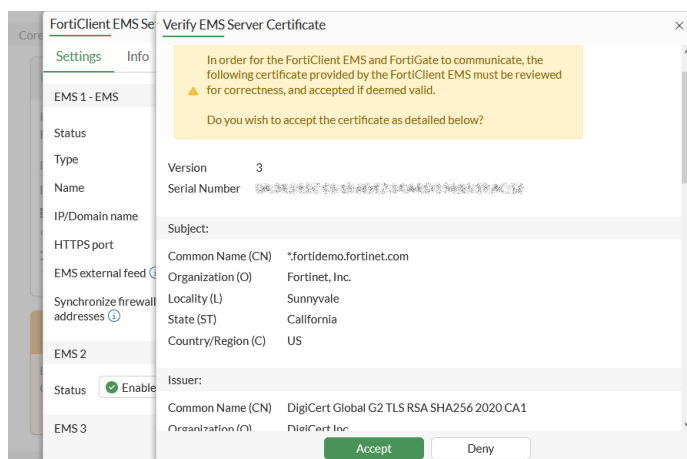
To add an on-premise FortiClient EMS server to the Security Fabric in the GUI:

1. Go to *Security Fabric > Fabric Connectors* and double-click the *FortiClient EMS* card.
2. Enable an EMS, and set *Type* to *FortiClient EMS*.
3. Enter a name and IP address or FQDN.

When connecting to a multitenancy-enabled EMS, Fabric connectors must use an FQDN to connect to EMS, where the FQDN hostname matches a site name in EMS (including "Default"). The following are examples of FQDNs to provide when configuring the connector to connect to the default site and to a site named SiteA, respectively: default.ems.yourcompany.com, sitea.ems.yourcompany.com. See [Multitenancy](#).

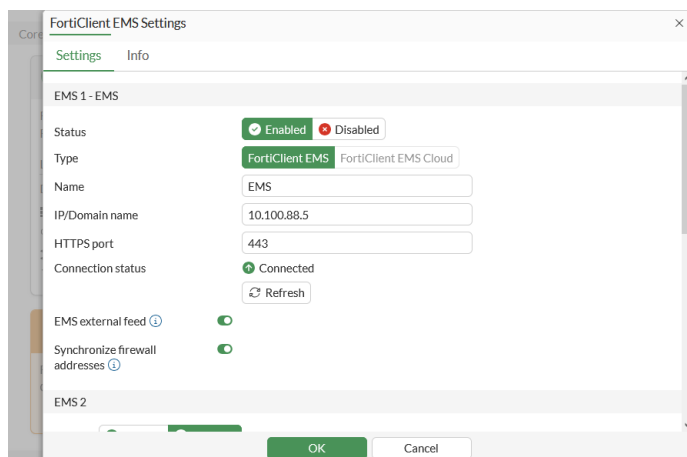
4. Optionally, enable *EMS External Feed*. See [Malware external feed from EMS on page 2412](#) for more information about using this setting in an AV profile.
5. Click *OK*.

A window appears to verify the EMS server certificate:



6. Click **Accept**.

The *Connection status* is now *Connected*:



7. If the device is not authorized, log in to the FortiClient EMS to authorize the FortiGate under *Administration > Fabric Devices*.

In FortiClient 7.2.5 and above, approvals are performed under *Fabric & Connectors > Fabric Devices*.

To add a FortiClient EMS Cloud server to the Security Fabric in the GUI:



FortiClient EMS Cloud can only be configured when the FortiGate is registered to FortiCloud and the EMS Cloud entitlement is verified.

If the FortiCloud account does not pass the FortiClient EMS Cloud entitlement check, the option is not selectable in the FortiClient EMS connector settings.

1. Go to *Security Fabric > Fabric Connectors* and double-click the *FortiClient EMS* card.
2. Set the *Type* to *FortiClient EMS Cloud*.
3. Enter a name.
4. Click **OK**.

A window appears to verify the EMS server certificate.

5. Click **Accept**.

The *Connection status* is now *Connected*.

To test connectivity with the EMS server:

1. Go to *Security Fabric > Fabric Connectors* and double-click the *FortiClient EMS* card.
2. In the *Connection status* section, click *Refresh*.

To add an on-premise FortiClient EMS server to the Security Fabric in the CLI:

```
config endpoint-control fctems
  edit {1 | 2 | 3 | 4 | 5}
    set status {enable | disable}
    set name <string>
    set server <ip_address>
    set certificate <string>
    set https-port <integer>
    set source-ip <ip_address>
  next
end
```

The https-port is the EMS HTTPS access port number, and the source-ip is the REST API call source IP address.

To add a FortiClient EMS Cloud server to the Security Fabric in the CLI:

```
config endpoint-control fctems
  edit {1 | 2 | 3 | 4 | 5}
    set status {enable | disable}
    set name <string>
    set fortinetone-cloud-authentication enable
    set certificate <string>
  next
end
```

To verify the EMS Cloud entitlement in the CLI:

```
# diagnose test update info
```

To verify an EMS certificate in the CLI:

```
# execute fctems verify ems137

Subject:      C = CA, ST = bc, L = burnaby, O = devqa, OU = top3, CN =
sys169.qa.fortinet.cm, emailAddress = xxxx@xxxxxxxxx.xxx
Issuer:       CN = 155-sub1.fortinet.com
Valid from:   2017-12-05 00:37:57 GMT
Valid to:     2027-12-02 18:08:13 GMT
Fingerprint: D3:7A:1B:84:CC:B7:5C:F0:A5:73:3D:BB:ED:21:F2:E0
Root CA:      No
Version:      3
Serial Num:   01:86:a2
```

Extensions:

Name: X509v3 Basic Constraints

Critical: yes

Content:

CA:FALSE

Name: X509v3 Subject Key Identifier

Critical: no

Content:

35:B0:E2:62:AF:9A:7A:E6:A6:8E:AD:CB:A4:CF:4D:7A:DE:27:39:A4

Name: X509v3 Authority Key Identifier

Critical: no

Content:

keyid:66:54:0F:78:78:91:F2:E4:08:BB:80:2C:F6:BC:01:8E:3F:47:43:B1

DirName:/C=CA/ST=bc/L=burnaby/O=devqa/OU=top3/CN=fac155.fortinet.com/emailAddress=xyguo@fortinet.com

serial:01:86:A4

Name: X509v3 Subject Alternative Name

Critical: no

Content:

DNS:sys169.qa.fortinet.cm

Name: X509v3 Key Usage

Critical: no

Content:

Digital Signature, Non Repudiation, Key Encipherment, Data Encipherment, Key Agreement, Certificate Sign, CRL Sign, Encipher Only, Decipher Only

Name: X509v3 Extended Key Usage

Critical: no

Content:

TLS Web Server Authentication, TLS Web Client Authentication

EMS configuration needs user to confirm server certificate.

Do you wish to add the above certificate to trusted remote certificates? (y/n)y

FortiClient multi-tenancy

Multi-tenancy gives administrators the flexibility to deploy a single FortiGate with access to multiple FortiClient EMS servers, or a single FortiClient EMS with multiple tenants. The FortiGate can support up to seven EMS servers in a single VDOM. When multi-VDOM is enabled on the FortiGate, each VDOM can override the global EMS configurations to connect to their own EMS servers.



The override feature requires FortiClient EMS 7.2.1 and later, and FortiGate running FOS 7.4.0 or later. To use override with FortiClient EMS Cloud, a FortiGate must be running FOS 7.4.4 or later. To connect to a FortiClient Cloud instance registered under a FortiCloud sub-OU in the GUI, a FortiGate must be running FOS 7.6.1 or later.

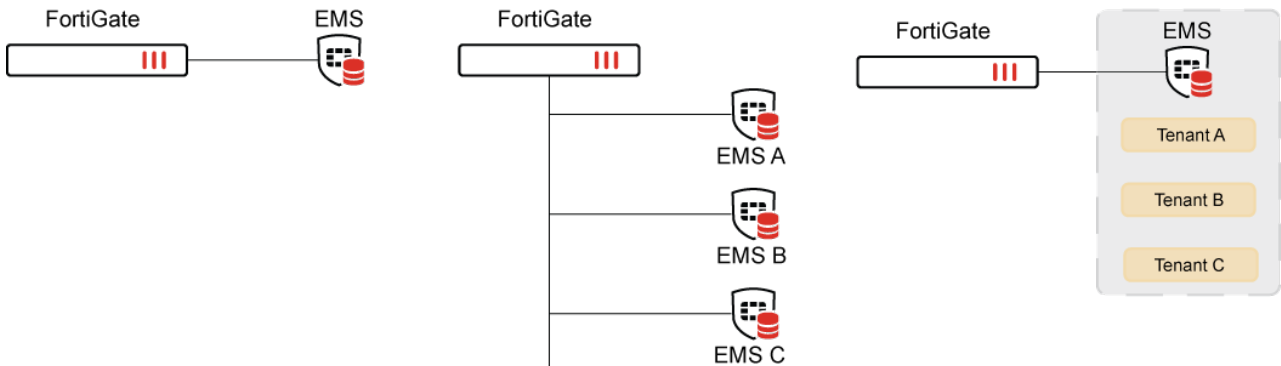
This functionality can be applied to MSSP (managed security service provider) configurations, and each VDOM has its own *FortiClient EMS* card for the EMS server or instance.

The following reference table provides a high-level view of single versus multi-tenancy scenarios, depending on the status of the FortiGate (whether VDOM is enabled or disabled) and FortiClient EMS:

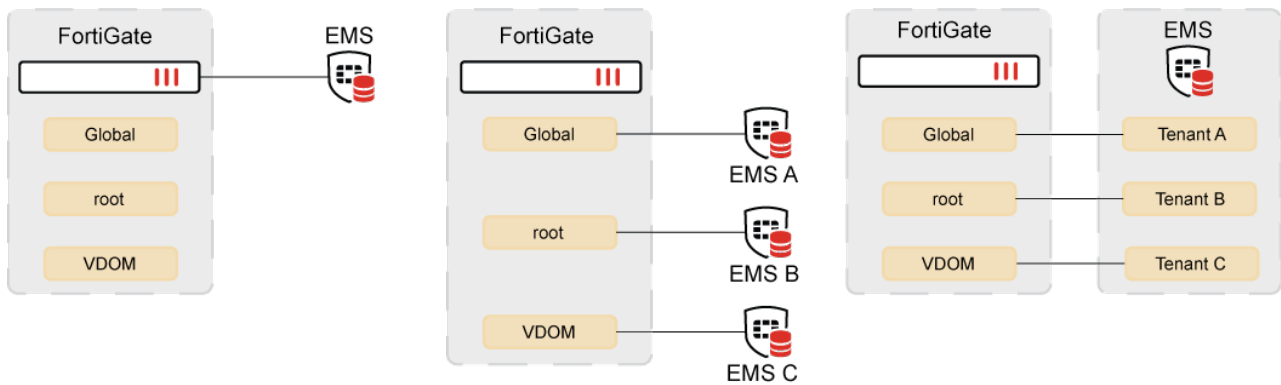
FortiGate	Single FortiClient EMS instance	Multiple FortiClient EMS instances	Single FortiClient EMS instance with multi-tenancy
No VDOM	Single tenant	Multi-tenancy without override	Multi-tenancy without override
VDOM	Global FortiClient EMS multi-tenancy	Multi-tenancy with override	Multi-tenancy with override

These scenarios can further be demonstrated as follows:

- No VDOM:



- VDOM:



Basic configurations

In a single VDOM configuration, you can configure multiple EMS servers.

To configure FortiClient EMS servers in a single VDOM set up:

1. Go to *Security Fabric > Fabric Connectors*.
2. Double-click on the *FortiClient EMS* card to edit.
3. For each EMS server, click *Enabled*, and fill in the configurations for that EMS server.
4. Click *OK* to save the settings.

In a multi-VDOM configuration, first configure the global EMS configurations, then configure override on each VDOM. If a VDOM does not enable override, it will inherit the global configurations.

To configure FortiClient EMS servers in a multi-VDOM set up:

1. In the Global VDOM, go to *Security Fabric > Fabric Connectors*.
2. Double-click on the *FortiClient EMS* card to edit.
3. For each EMS server, click *Enabled*, and fill in the configurations for that EMS server.
4. Click *OK* to save the settings.
5. Enter a VDOM.
6. From the CLI, edit the following settings:

```
config endpoint-control settings
    set override enable
end
```

7. Back in the GUI, go to *Security Fabric > Fabric Connectors*.
8. Configure each EMS server as needed.
9. Click *OK* to save.

Advanced configurations

FortiGate supports connecting to a FortiClient Cloud instance registered under a sub-OU in FortiCloud. Furthermore, a FortiGate can override FortiClient Cloud access key setting on a per-VDOM basis. With these enhancements, a FortiGate can support FortiClient Cloud in multi-tenancy scenarios.



This feature includes the following scope and limitations:

- The FortiGate will perform an entitlement check on the registered FortiCloud Account to verify a FortiClient Cloud entitlement exists on the root FortiCloud account. If the FortiGate has no FortiClient Cloud entitlement, you cannot select the FortiClient EMS Cloud type or input an access key.
- Using the FortiClient Cloud access key, a FortiGate can connect to a FortiClient Cloud instance belonging to a sub-OU in the same FortiCloud account or a different FortiCloud account.
- Within the same VDOM, the FortiGate can have an EMS connector connecting to multiple FortiClient Cloud instances.

The FortiClient Cloud access key can be implemented in the `cloud-authentication-access-key` parameter in the CLI.

```
config endpoint-control fctems-override
  edit 1
    set status enable
    set name <name>
    set fortinetone-cloud-authentication enable
    set cloud-authentication-access-key <key>
  next
end
```

Examples

Example 1: Enabling override on the root VDOM using the CLI

To enable override on the root VDOM in the CLI:

1. Enable override on the required VDOMs:

```
config endpoint-control settings
  set override enable
end
```

2. Configure the EMS server on the desired VDOM:

```
(root) config endpoint-control fctems-override
  edit 1
    set status enable
    set name "ems140_root"
    set server "172.16.200.140"
    set serial-number "FCTEMS8821*****"
    set tenant-id "00000000000000000000000000000000"
    set capabilities fabric-auth silent-approval websocket websocket-malware push-ca-certs
  common-tags-api tenant-id single-vm-domain-connector
  next
  edit 2
    set name "ems133_root"
    set server "172.16.200.133"
  next
end
```

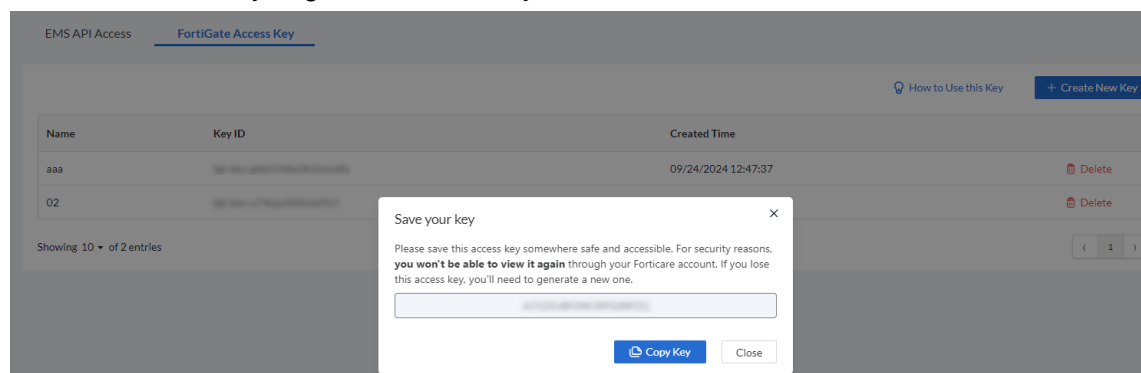
Example 2: Connecting to different FortiClient Cloud instances

In this example, a FortiGate will connect to different FortiClient Cloud instances between the Global EMS connector, root and vdom1.

To connect to different FortiClient Cloud instances in the CLI:

1. Obtain the access by from FortiClient Cloud by going to *FortiCloud > FortiClient Cloud*.
2. Click *Access Key* and switch to the *FortiGate Access Key* tab.

3. Click *Create New Key* to generate a new key.



4. Repeat this for another FortiClient Cloud instance to be applied to vdom1.
5. On the FortiGate with multi-VDOM enabled, configure the Global EMS connector:

```
config global
  config endpoint-control fctems
    edit 2
      set status enable
      set name "Cloud_EMS_Global"
      set fortinetone-cloud-authentication enable
      set serial-number "FCTEMSXXXXXXXXXX"
      set tenant-id "00000000000000000000000000000000"
    next
  end
end
```

6. Switch to and configure the root VDOM:

```
config vdom
  edit root
    config endpoint-control settings
      set override enable
    end
    config endpoint-control fctems-override
      edit 1
        set status enable
        set name "cloud_ems_root"
        set fortinetone-cloud-authentication enable
        set cloud-authentication-access-key "XXXXXXXXXXXXXXXXXXXX"
        set serial-number "FCTEMSXXXXXXXXXX"
        set tenant-id "00000000000000000000000000000000"
      next
    end
  next
end
```

7. Repeat the same steps for vdom1:

```
config vdom
  edit vdom1
```

```

config endpoint-control settings
    set override enable
end
config endpoint-control fctems-override
    edit 1
        set status enable
        set name "cloud_vdom1"
        set fortinetone-cloud-authentication enable
        set cloud-authentication-access-key "XXXXXXXXXXXXXXXXXXXX"
        set serial-number "FCTEMSXXXXXXXXXX"
        set tenant-id "00000000000000000000000000000000"
    next
end
next
end

```

8. From the CLI, run the following commands to troubleshoot.

```

# diagnose endpoint filter show-large-data yes
# diagnose debug application fcnacd -1
# diagnose debug enable

```

A successful connection will look like the following:

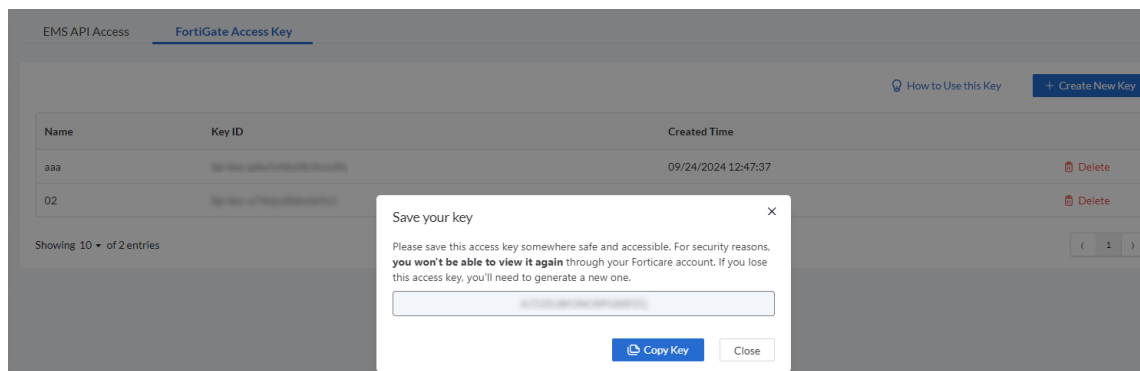
```

...
[ec_ez_worker_base_prep_resolver:382] Outgoing interface index 0 for 1 (cloud_vdom1).
[ec_ez_worker_prep_data_url:190] Full URL: https://sf.00000-XXXXXXXXXXXXXXXXXXXX.fortinet-
ca2.fortinet.com/api/v1/system/serial_number
[ec_ez_worker_base_prep_ssl:429] verify peer method: 3, current ssl_cb: (nil), new ssl_cb:
0x55c1163571b0
[ec_ems_context_submit_work:642] Call submitted successfully.
    obj-id: 0, desc: REST API to get EMS Serial Number., entry: api/v1/system/serial_number.
[__match_server_cert_key:462] verify_peer_method: 3

```

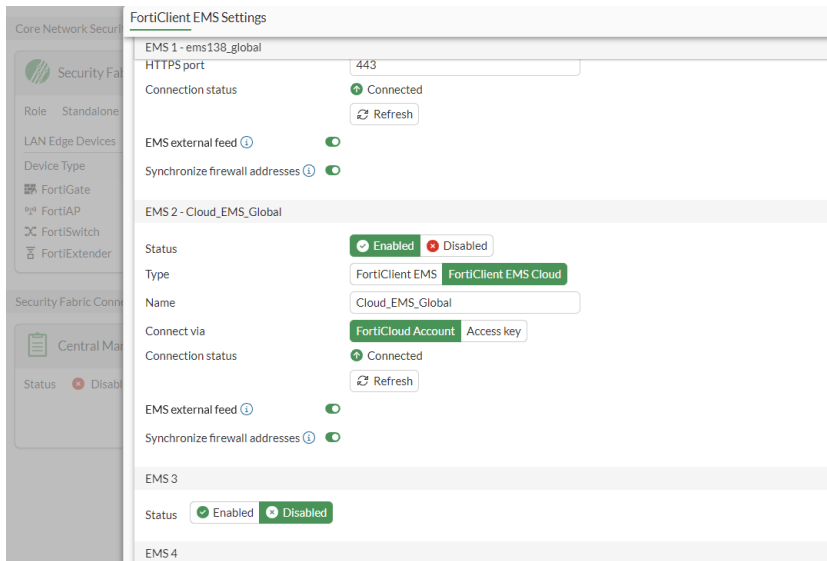
To connect to different FortiClient Cloud instances in the GUI:

1. Obtain the access by from FortiClient Cloud by going to *FortiCloud > FortiClient Cloud*.
2. Click *Access Key* and switch to the *FortiGate Access Key* tab.
3. Click *Create New Key* to generate a new key.

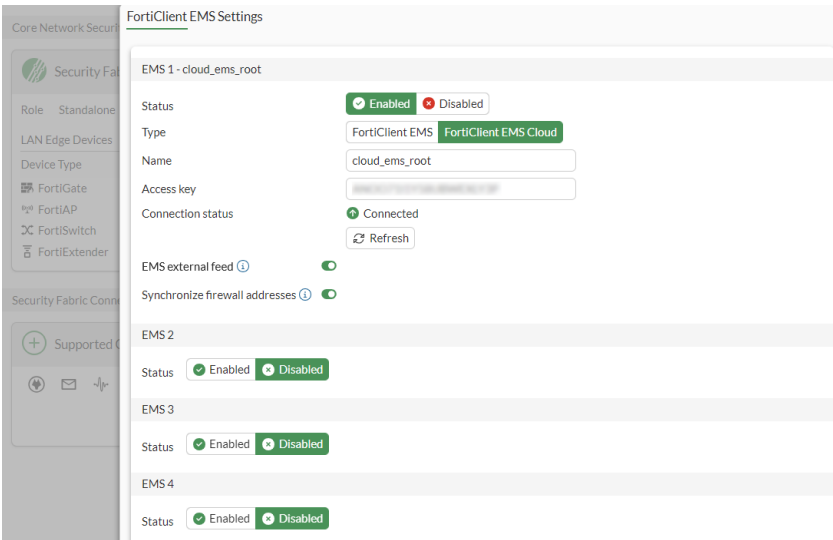


4. Repeat this for another FortiClient Cloud instance to be applied to vdom1.

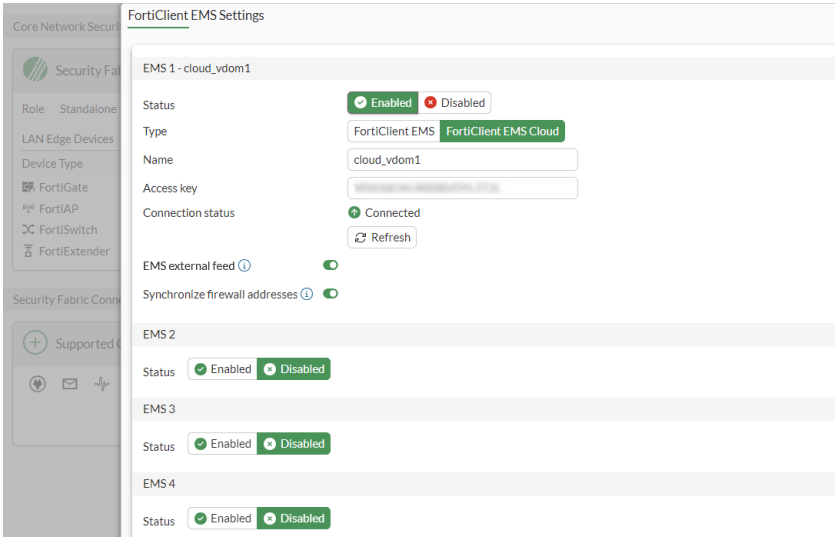
5. On the FortiGate with multi-vdom enabled, go to *Global*.
 - a. Go to *Security Fabric > Fabric Connectors*.
 - b. Edit the *FortiClient EMS* connector.
 - c. Set *Status* to *Enabled*.
 - d. Set *Type* to *FortiClient EMS Cloud*.
 - e. Set *Name* to *Cloud_EMS_Global*.
 - f. Set *Connect via* to *FortiCloud Account*.
 - g. Click *OK* to save. Verify the certificate when prompted and continue saving the settings.



6. Switch to the root vdom.
 - a. Go to *Security Fabric > Fabric Connectors*.
 - b. Edit the *FortiClient EMS* connector.
 - c. Set *Status* to *Enabled*.
 - d. Set *Type* to *FortiClient EMS Cloud*.
 - e. Set *Name* to *cloud_ems_root*.
 - f. Set *Access key* to the key retrieved from FortiClient Cloud.
 - g. Click *OK* to save.



7. Repeat the same steps above for vdom1.



FortiClient EMS capabilities

FortiClient EMS supports many capabilities that are integrated with the FortiGate though the EMS connector. New versions of FortiClient EMS may support new capabilities. The FortiGate can detect and synchronize the capabilities from each EMS server as it establishes the EMS connection.

Use the `config endpoint-control fctems` command on FortiGate to enable the following capabilities:

Capability	Description
fabric-auth	Allow this FortiGate to load the authentication page provided by EMS to authenticate itself with EMS.
silent-approval	Allow silent approval of non-root or FortiGate HA clusters on EMS in the Security Fabric.

Capability	Description
	For an example, see Using EMS silent approval in the Security Fabric on page 4553 .
websocket	Enable/disable websockets for this FortiGate. Override behavior using websocket-override.
websocket-malware	Allow this FortiGate to request malware hash notifications over websocket.
push-ca-certs	Enable/disable syncing deep inspection certificates with EMS. For an example, see Allowing deep inspection certificates to be synchronized to EMS and distributed to FortiClient on page 4554 .
common-tags-api	Can receive tag information from New Common Tags API from EMS.
tenant-id	Allow this FortiGate to retrieve Tenant-ID from EMS.
client-avatars	Allow this FortiGate to retrieve avatars from EMS by fingerprint.
single-vdom-connector	Allow this FortiGate to create a VDOM connector to EMS.
fgt-sysinfo-api	Allow this FortiGate to send additional info to EMS.
zttna-server-info	Allow this FortiGate to send VDOM's ZTNA server information to EMS.
used-tags	Allow this FortiGate to send used-tag information to EMS through the FortiOS REST API. Requires FortiClient EMS 7.4.4 or later. With the used-tags capability set, FortiGate uses the REST API (fabric/tag_list) to: • Send the updated used-tag list to EMS when a tag is added, updated, or removed from FortiGate • Send an empty used-tag list to EMS when a VDOM is added or deleted on FortiGate • Send all used tags from all VDOMs on FortiGate • Send all used tags to EMS after a connection between FortiGate and EMS is restored after being down.

Using EMS silent approval in the Security Fabric

FortiClient EMS with Fabric authorization and silent approval capabilities can approve the root FortiGate in a Security Fabric once, and then silently approve remaining downstream FortiGates in the Fabric. Similarly in an HA scenario, an approval only needs to be made once to the HA primary unit. The remaining cluster members are approved silently.

To use EMS silent approval:

1. Configure the EMS entry on the root FortiGate or HA primary:

```
config endpoint-control fctems
  edit 1
    set status enable
```

```

    set name "ems139"
    set fortinetone-cloud-authentication disable
    set server "172.16.200.139"
    set https-port 443
    set source-ip 0.0.0.0
    set pull-sysinfo enable
    set pull-vulnerabilities enable
    set pull-tags enable
    set pull-malware-hash enable
    unset capabilities
    set call-timeout 30
    set websocket-override disable
  next
end

```

When the entry is created, the capabilities are unset by default.

2. Authenticate the FortiGate with EMS:

```

# execute fctems verify ems_139
...

```

The FortiGate will enable the Fabric authorization and silent approval based on the EMS supported capabilities.

```

config endpoint-control fctems
  edit 1
    set server "172.18.62.12"
    set capabilities fabric-auth silent-approval websocket
  next
end

```

3. Configure a downstream device in the Security Fabric (see [Configuring the root FortiGate and downstream FortiGates on page 4520](#) for more details). The downstream device will be silently approved.
4. Configure a secondary device in an HA system (see [HA active-passive cluster setup on page 4284](#) and [HA active-active cluster setup on page 4287](#) for more details). The secondary device will be silently approved.

Allowing deep inspection certificates to be synchronized to EMS and distributed to FortiClient

On FortiClient EMS versions that support push CA certs capability, the FortiGate will push CA certificates used in SSL deep inspection (see [Deep inspection on page 2850](#) for more details) to the EMS server. On the EMS server, the CA certificates can be selected in the managed endpoint profiles so they can be installed on managed endpoints. FortiClient EMS 7.0.1 and later is required to use this feature.

To configure deep inspection certificate synchronization to EMS:

1. Configure the EMS Fabric connector:

```

config endpoint-control fctems
  edit 2
    set status enable

```

```

    set name "ems138"
    set fortinetone-cloud-authentication disable
    set server "172.16.200.138"
    set https-port 443
    set source-ip 0.0.0.0
    set pull-sysinfo enable
    set pull-vulnerabilities enable
    set pull-tags enable
    set pull-malware-hash enable
    set capabilities fabric-auth silent-approval websocket websocket-malware push-ca-certs
    set call-timeout 30
    set websocket-override disable
  next
end

```

2. Apply the certificate to an SSL/SSH profile for deep inspection:

```

config firewall ssl-ssh-profile
  edit "deep-inspection"
    set comment "Read-only deep inspection profile."
    config https
      set ports 443
      set status deep-inspection
    end
    ...
    set server-cert-mode re-sign
    set caname "Fortinet_CA_SSL"
    set untrusted-caname "Fortinet_CA_Untrusted"
  next
end

```

The default deep inspection profile, CA certificate, and untrusted CA certificates are used in this example.

3. Configure the firewall policy:

```

config firewall policy
  edit 1
    set name "deep-inspection"
    set srcintf "port14"
    set dstintf "port13"
    set action accept
    set srcaddr "all"
    set dstaddr "all"
    set schedule "always"
    set service "ALL"
    set utm-status enable
    set inspection-mode proxy
    set ssl-ssh-profile "deep-inspection"
    set av-profile "default"
    set nat enable
  next
end

```

4. In EMS, verify that the CA certificate was pushed to EMS:

a. Go to *Endpoint Policy & Components > CA Certificates*.

Name	Subject	Expiry
FGVM02TM20011370[...]		
Fortinet_CA_SSL	/C=US/ST=California/L=Sunnyvale/O=Fortinet/OU=Certificate Authority/CN=FGVM02TM20011370/e...	2031-07-26 18:16:52
Fortinet_CA_Untrusted	/C=US/ST=California/L=Sunnyvale/O=Fortinet/OU=Certificate Authority/CN=Fortinet Untrusted CA/e...	2031-07-26 18:15:08
FGVM02TM21012631[...]		
Fortinet_CA_SSL	/C=US/ST=California/L=Sunnyvale/O=Fortinet/OU=Certificate Authority/CN=FGVM02TM21012631/e...	2031-08-11 17:00:15
Fortinet_CA_Untrusted	/C=US/ST=California/L=Sunnyvale/O=Fortinet/OU=Certificate Authority/CN=Fortinet Untrusted CA/e...	2031-08-11 16:58:17
FGVM02TM21011327[...]		
Fortinet_CA_SSL	/C=US/ST=California/L=Sunnyvale/O=Fortinet/OU=Certificate Authority/CN=FGVM02TM21011327/e...	2031-08-11 16:59:33
Fortinet_CA_Untrusted	/C=US/ST=California/L=Sunnyvale/O=Fortinet/OU=Certificate Authority/CN=Fortinet Untrusted CA/e...	2031-08-11 16:58:18
FGVM02TM21010811[...]		
Fortinet_CA_SSL	/C=US/ST=California/L=Sunnyvale/O=Fortinet/OU=Certificate Authority/CN=FGVM02TM21010811/e...	2031-08-11 16:59:33
Fortinet_CA_Untrusted	/C=US/ST=California/L=Sunnyvale/O=Fortinet/OU=Certificate Authority/CN=Fortinet Untrusted CA/e...	2031-08-11 16:58:18
FGVM02TM21012111[...]		
Fortinet_CA_SSL	/C=US/ST=California/L=Sunnyvale/O=Fortinet/OU=Certificate Authority/CN=FGVM02TM21012111/e...	2031-08-11 17:00:14

b. Verify the certificate table to see that the EMS server received the CA certification from the different FortiGates.

5. Select the CA certificate in the endpoint profile:

- Go to *Endpoint Profiles > System Settings* and edit a profile. The default profile is used in this example.
- In the *Other* section, enable *Install CA Certificate on Client* and select the *Fortinet_CA_SSL* certificate for the desired endpoint.

System Settings Profile

Expand All Collapse All

Name: Default Basic Advanced XML

☐ Fortinet_CA_SSL Valid
☐ Fortinet_CA_Untrusted Valid
 Imported from FGVM02TM20011370
☒ Fortinet_CA_SSL Valid until 2031-08-12 00:00:15
☐ Fortinet_CA_Untrusted Valid
 Imported from FGVM02TM21012631
☐ Fortinet_CA_SSL Valid
☐ Fortinet_CA_Untrusted Valid
 Imported from FGVM02TM21011327
☐ Fortinet_CA_SSL Valid
☐ Fortinet_CA_Untrusted Valid
 Imported from FGVM02TM21010811
☐ Fortinet_CA_SSL Valid
☐ Fortinet_CA_Untrusted Valid
 Imported from FGVM02TM21012111
☐ Fortinet_CA_SSL Valid

Save Discard Changes Revert To Default

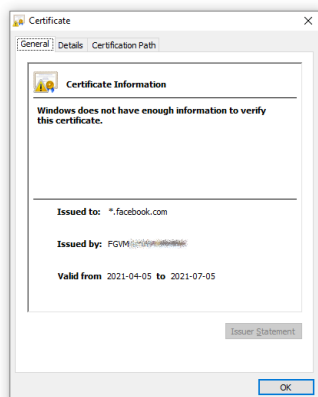
c. Click Save.

After the FortiClient endpoint is registered, it receives the CA certificate. When the FortiClient endpoint tries to access the internet through the FortiGate with the firewall policy that has deep inspection, no warning message is displayed. The server certificate is trusted with the installed CA certificate to complete the certificate chain.

Verification

Before configuring deep inspection certificate synchronization, a warning message is displayed when a FortiClient endpoint accesses the internet through the FortiGate with the firewall policy that has deep inspection. The FortiClient certificate store does not have the FortiGate's CA that is used in the deep inspection SSL/SSH profile.

For example, accessing <https://www.facebook.com> in Chrome shows a warning. In the address bar, clicking *Not secure > Certificate* opens the *Certificate* dialog, which indicates that *Windows does not have enough information to verify the certificate*.



Chrome from safely

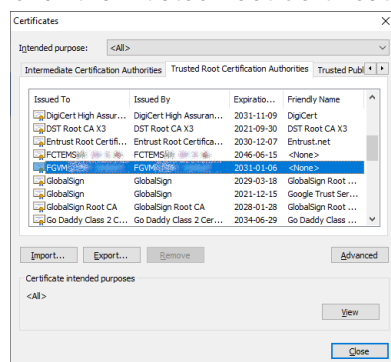
computer or network. Ask your IT

urity, [turn on enhanced protection](#)

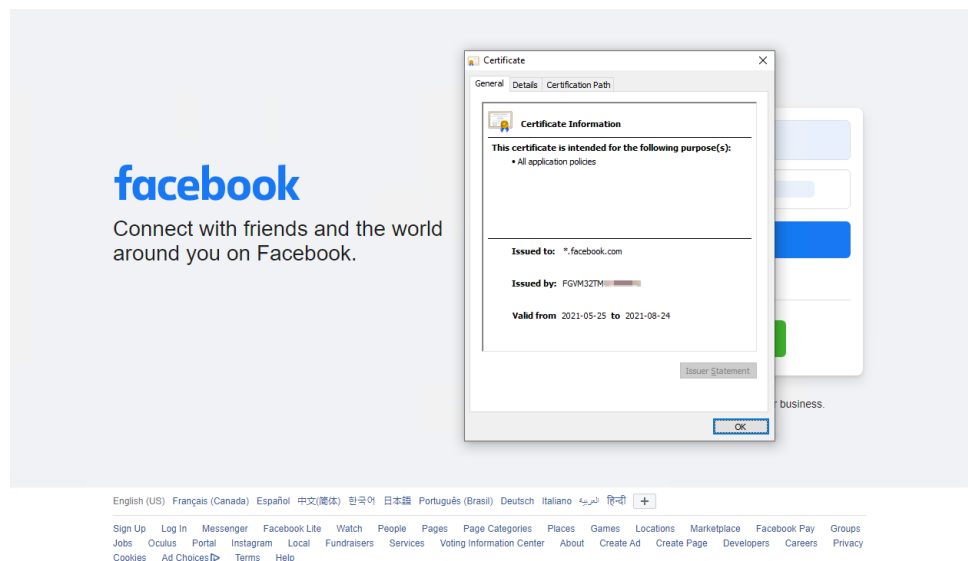
After the EMS profile is pushed to FortiClient endpoint, the expected FortiGate's certificate is shown in its certificate store.

To verify the deep inspection certificate synchronization:

1. In Chrome, go to *Settings > Privacy and security* and open *Manage certificates*.
2. Click the *Trusted Root Certification Authorities* tab. The FortiGate's certificate appears in the list.



3. On the FortiClient endpoint using Chrome, go to <https://www.facebook.com>. The website is displayed.
4. In the address bar, click the padlock, then click *Certificate*. The dialog displays the valid certificate information.



Diagnostics

Use the `diagnose endpoint fctems json deep-inspect-cert-sync` command in FortiOS to verify the certificate information. In the following example, there are multiple VDOMs with FortiGates in HA mode.

To verify the primary FortiGate:

```
FGT_EC_Primary (global) # diagnose endpoint fctems json deep-inspect-cert-sync
JSON:
"""
{
  "fortigates":[
    "FG2K5E39169*****",
    "FG2K5E39169*****"
  ],
  "vdoms":[
    {
      "vdom":"root",
      "certs":[
        {
          "name":"Fortinet_CA_SSL",
          "cert":"-----BEGIN CERTIFICATE-----\\nMIID5jCCAs6g...Sfu+Q8zE8Crmt6L1X\\n/bv+q\\n-----END
CERTIFICATE-----\\n"
        },
        {
          "name":"Fortinet_CA_Untrusted",
          "cert":"-----BEGIN CERTIFICATE-----\\nMIID8DCCAtig...3zBbfzP+nVUPC\\nZDPRZA==\\n-----END
CERTIFICATE-----"
        }
      ]
    },
    {
      "vdom":"vdom1",
```